Rupam Kumar

Artificial Intelligence | Machine Learning | Generative AI

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LinkedIn | GitHub

Professional Summary

Mechanical Engineering graduate from NIT Durgapur with project experience in retrieval-augmented generation, semantic retrieval, transformer embeddings, and AI application development. Built a context-grounded conversational guide with a FastAPI service and response validation. Familiar with Python, scikit-learn, TensorFlow, NLP, and core machine-learning workflows; interested in AI/ML and Generative AI engineering opportunities.

Technical Skills

Programming: Python, SQL, C++
Machine Learning: Scikit-learn, TensorFlow, Regression, Classification, Clustering, Feature Engineering, Model Evaluation
NLP & Transformers: Natural Language Processing, Transformer Embeddings
Generative AI: LLMs, Retrieval-Augmented Generation (RAG), Semantic Search, Vector Embeddings, Prompt Engineering
Application Development: FastAPI, REST APIs
Data Tools: Pandas, NumPy, Matplotlib, Seaborn, Excel

Professional Experience

Reliance Industries Limited (RIL) Sep 2026 – Present
Graduate Engineer Trainee – CES Rotary Nagothane, Maharashtra

- Review rotating equipment parameters, including vibration and temperature readings, to support equipment condition assessment and performance monitoring.
- Examine operating trends and document observations to support abnormality identification and equipment reliability.

GPB Caliberidx Private Limited May 2025 – Jul 2025
Data Analyst Intern Gaya, Bihar

- Developed 5 Power BI dashboards for 15+ stakeholders, improving KPI visibility and business reporting.
- Streamlined recurring reporting using SQL and Python, saving 8+ hours of manual effort per week.
- Analyzed 100K+ transactional records to investigate revenue leakage and customer retention gaps.

Projects

GitaGuideAI – AI Conversational Guide Python, FastAPI, Llama API

- Developed a retrieval-augmented generation (RAG) pipeline combining semantic retrieval and transformer embeddings to generate context-grounded responses.
- Built a FastAPI REST service with prompt augmentation and response validation for structured outputs.

Vriksh Students Federation Portal Next.js, Prisma, PostgreSQL, Cloudinary, Vercel

- Built and deployed a full-stack portal centralizing student records, monthly contributions, and administrative workflows.
- Implemented protected document access, private file storage, audit logging, and transactional email workflows.
- **Live:** vriksh-sf.vercel.app

JEsquare – Online Assessment Platform In Progress

- Developing an online assessment platform for students to conduct tests and streamline evaluation, with OCR integration for automated answer extraction and processing.

Education

National Institute of Technology Durgapur 2022 – May 2026
B.Tech in Mechanical Engineering CGPA: 6.95/10

Pragya Bharti Public School, Gaya, Bihar
Senior Secondary: 89% (2022) | Secondary: 90% (2020)

Leadership & Achievements

Chairperson, ISTE Students' Chapter, NIT Durgapur May 2025 – May 2026
Led a 35+ member team and coordinated technical events for 500+ participants.

Secretary, Vriksh Students Federation Sep 2021 – Present
Supported educational and mentorship initiatives for 200+ students.

Achievement: Qualified JEE Advanced.

Certifications

Machine Learning A-Z – Udemy, 2025 | Complete Data Analyst Bootcamp – Udemy, 2025